MSc. in Computing
Practicum Approval Form

Section 1: Student Details

Project Title:	Using Electroencephalography and Inverse Reinforcement Learning to measure impulsivity in order to detect and diagnose attention deficit hyperactivity disorder in users.
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Chosen major:	Artificial Intelligence
Supervisor	Tomas Ward
Date of Submission	20/11/2025

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

Using Elelctroencephalography to record brain signals and then using Inverse Reinforcement Learning to measure a person's impulsivity in order to potentially detect and even diagnose them with attention deficit hyperactivity disorder. The project involves the user performing a behavioural task. This task will be in the form of a serious game, specifically a Highway task, where the user has to drive safely and avoid oncoming traffic.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. **Bridging the Gap Between Self-Report and Behavioral Laboratory Measures: A Real-Time Driving Task with Inverse Reinforcement Learning** (by Sang Ho Lee, Myeong Seop Song, Min-hwan Oh, and Woo-Young Ahn):
    a. The user played a Highway task in which they would dodge cars and make decisions (by changing lanes for example) in order to drive safely. If the user crashed too many times, and their fuel ran out, then the task would end.

2. **Modeling sensory-motor decisions in natural behavior** (by Ruohan Zhang, Shun Zhang, Matthew H. Tong, Yuchen Cui, Constantin A. Rothkopf, Dana H. Ballard, Mary M. Hayhoe)
    a. In this paper, they aim to prove that an inverse reinforcement model can be developed in order to predict the rewards from human behavioural data.
3. **fNIRS-EEG BCIs for Motor Rehabilitation: A Review** (by Jianan Chen, Yunjia Xia, Xinkai Zhou, Ernesto Vidal Rosas, Alexander Thomas, Rui Loureiro, Robert J. Cooper, Tom Carlson and Hubin Zhao)
    a. This paper details the usefulness of Electroencephalograms and Brain Computer Interfaces for motor rehabilitation by recording brain signals and using that data to develop Brain Computer Interface applications that control some application.
4. **Virtual Reality for Enhanced Ecological Validity and Experimental Control in the Clinical, Affective and Social Neurosciences** (by Thomas D. Parsons)
    a. This is about Using Virtual Reality to simulate real world conditions. This paper in particular mentions one researcher, who dismisses the claims of another researcher, on the basis that their experiment isn't realistic or representative of the real world.
5. **Neurophysiological methods for monitoring brain activity in serious games and virtual environments: a review** (by Manuel Ninaus, Silvia Erika Kober, Elisabeth V.C. Friedrich, Ian Dunwell, Sara de Freitas, Sylvester Arnab, Michela Ott, Milos Kravcik, Theodore Lim, Sandy Louchart, Francesco Bellotti, Anna Hannemann, Alasdair G. Thin, Riccardo Berta, Guilherme Wood, Christa Neuper)
    a. A review of neuroscientific methods and techniques used in conjunction with serious games / game-based learning.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Our project builds upon the ideas from the first paper. However, we differ in the type of task that is being used, which is a highly realistic serious game, where the user's impulsivity is measured during one playthrough of the task. This is to address the potentially flawed ecological validity of the first paper. We also incorporate Electroencephalography with inverse reinforcement learning in order to get a real time reading of the user's impulsivity and ultimately detect or diagnose ADHD. The second paper details how inverse reinforcement learning can predict the rewards that someone values from their behavioural data. The third paper showcases why Electroencephalograms are useful, as well as their advantages and disadvantages. The fourth paper is about the importance of realistic experiments, and how virtual reality can be used to immerse the user for more accurate decision making and experiments. The last paper is a general review of different methods used with serious games and neuroscience techniques.

What are the research questions that you will attempt to answer? (200 words)

Can we detect attention deficit hyperactivity disorder in people? Traditionally, ADHD is difficult to detect, and a series of questionnaires, interviews and tests usually take place in order to diagnose someone with ADHD. If it is possible to detect and diagnose ADHD, then this could potentially lead to an easier method of detecting other neurological disorders.

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?
    - Unreal Engine 5, Visual Studio with C++, OpenVibe for LSL.
- What coding/development will you do?
    - Develop the Highway task in C++ and Blueprint, build an Inverse Reinforcement Learning model, as well as develop a pipeline in order to record brain signals, and send them to the game, where the behaviour of a user will be displayed after the task.
- What data will be used for your investigations?
    - Since we are using an Electroencephalogram, it wouldn't be useful to use a dataset with random brain signal data, since we need the user to play this game while recording their brain signal data. Thus, our data will come from our own recordings while playing the game.
- Is this data currently available, it not, where will it come from?
    - Yes, the data will come from our own recordings while playing the game.
- What experiments do you expect to run?
    - Similar to the first paper, our experiment would be to run the task and then record the results of the level of impulsivity. Impulsivity is a core symptom of ADHD, so our experiment's goal is to possibly detect ADHD.
- What output do you expect to gather?
    - Our ourput would be a score of how impulsive a person is, a long with other results such as their score on the task, which can measure whether or not they could have ADHD.
- How will the results be evaluated?
    - The results will be evaluated by seeing how the accuracy of the impulsivity relates to traditional ADHD tests and scales, such as the Adult ADHD Self-Report Scale (ASRS).